Chapter 7
Force and Laws of Motion

***Force:** A force is an agent in the form of a push or pull which produces a change in the state of rest or motion of an object on which it acts.

Q. State three applications of force
Ans: The three applications of force are:

1. A force can be used to change the shape and size of an object.
2. It can be used to change the direction of an object in motion.
3. It can be used to change the velocity of an object.

*** Balanced Force:** If a force acting on an object produces no acceleration in it, then the force is said to be a balanced force.

*** Unbalanced Force:** If a force acting on an object produces an acceleration in it, then the force is said to be an unbalanced force.

LAWS OF MOTION:

*** Newton's First Law of Motion:** It states that every object continues in its state of rest or of motion unless an external unbalanced force acts on it.

Q. Why is Newton's first law known as law of inertia?
Ans: The tendency of an undisturbed object to remain at rest or to continue moving with the same velocity is called inertia. Because of this tendency, the first law is called law of inertia.

*** Inertia:** The inherent property of a body due to which it tends to continue in motion if the body is in motion and tends to continue at rest if the body is at rest is called inertia of a body.

Q. A passenger sitting in a moving bus moves forward when the bus stops suddenly. Give reason.
Ans: When the bus stops suddenly, the lower part of the body stops with the bus, but the upper part tends to keep moving due to inertia of motion. Therefore, the passenger falls forward.

Q. A passenger standing on a bus falls backward when the bus begins to start suddenly. Give reason.
Ans: This is because the sudden start of the bus brings motion to the bus as well as to the feet of the passenger that is in contact with the floor of the bus. But the rest of our body is still in a tendency of rest because of its inertia.

Q. Explain why some of the leaves may get detached from a tree if we vigorously shake its branch.
Ans: When we vigorously shake the branch of a tree, the branch comes into motion while the leaves on the branch tend to remain at rest due to inertia. That is why the leaves get detached from the branch when we shake it vigorously.

Q. When a hanging carpet is beaten with a stick, dust particles come out of it. Explain. Ans. The carpet moves suddenly when beaten, but dust particles remain at rest due to inertia and get separated.

*** Mass:** The mass of a body is the measure of its inertia and its SI unit is kg
OR The physical quantity which determines the inertia of a body is known as the mass of a body.

Note: More massive bodies will have larger inertia.

*** Momentum:** The momentum of a body is defined as the quantity of motion associated with the body.
OR, It is also defined as the product of mass and velocity of an object in motion.

i.e.

$$p = mv$$

*** SI Unit** of momentum (p) is $kgms^{-1}$.

*** Newton's Second Law of Motion** It states that the rate of change of momentum of a body is proportional to the applied unbalanced force in the direction of force.

*** Mathematical Formulation:**
Suppose an object of mass, m is moving along a straight line with an initial velocity, u. It is uniformly accelerated to velocity, v in time, t by the application of a constant force, F throughout the time, t . The initial and final momentum of the object are $p_1 = mu$ and $p_2 = mv$ respectively.

The change in momentum $\propto p_2 - p_1$

$$\propto mv - mu$$

$$\propto m(v - u)$$

The rate of change of momentum $\propto \dfrac{m \times (v-u)}{t}$

Or, The applied force,

$$F \propto \frac{m(v-u)}{t}$$

$$\Rightarrow F \propto \frac{mv-mu}{t}$$

$$\Rightarrow F = k\frac{m(v-u)}{t}$$

$$\Rightarrow F = kma$$

$$\Rightarrow F = ma \quad [\because k = 1]$$

*** SI Unit of Force** is Newton (N) or $kgms^{-2}$.

*** Definition of 1 Newton of force:** The force acting on a body of mass 1 kg is said to be 1 Newton if it produces an acceleration of 1 m s^{-2}.

$$1N = 1kg \times 1ms^{-2}$$

Q. Why does a fielder pull his hands backward while catching a cricket ball?
Ans. The fielder increases the time during which the ball is brought to rest. This decreases the rate of change of momentum and reduces the impact of force. Therefore, the catch becomes easier and safer.

***Impulse** The impulse of a force acting on a body is defined as the change in momentum of the body produced by the force. OR, It is also defined as the product of force and time.

$$I = Ft$$

*** SI Unit** of impulse is Ns or $kgms^{-1}$.

*** Impulse and Momentum:**

$$F = ma = m\frac{v-u}{t}$$

$$\Rightarrow Ft = mv - mu$$

$$\Rightarrow \text{Impulse} = \text{change in momentum}$$

*** Newton's Third Law of Motion:** It states that to every action, there is an equal and opposite reaction.

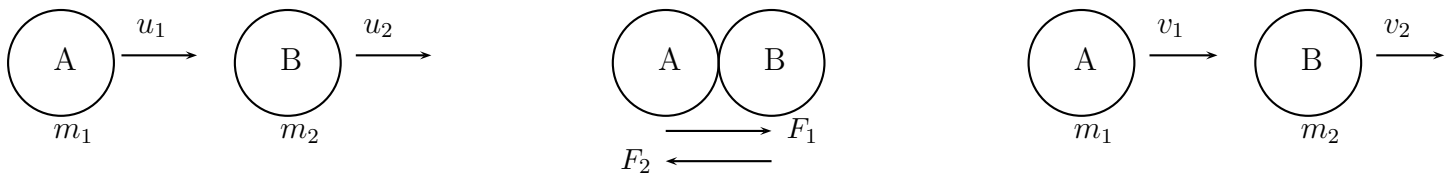
Action: The initial force exerted by one object onto another.

Reaction: The simultaneous force exerted by the second object back onto the first, equal in strength but opposite in direction.

Example: How can we walk on the ground? Explain on the basis of Newton's third law of motion.
Ans: We can walk on the ground on the basis of Newton's third law of motion in which our feet exert a force (action) on the ground, the ground also exerts an equal and opposite force (reaction) to our feet in order to move forward.

Q. State Law of Conservation of Momentum. Show that total momentum of two colliding bodies remains unchanged or conserved provides no external unbalanced force acts.
Ans: Law of Conservation of Momentum:

Momentum can neither be created nor destroyed - in a closed system consisting of a number of bodies colliding with one another the total momentum of the system remains constant



a) Before Collision b) During Collision c) After Collision

Consider two bodies of masses m_1 and m_2 moving along a straight line in the same direction at different velocities u_1 and u_2 such that $u_1 > u_2$ so that $m_1$and m_2 collide with each other as shown in figure.

From the figure,
Let F_1 = force exerted by body A on body B
F_2 = force exerted by body B on body A
m_1u_1, m_2u_2 = momenta of body A and B before collision
m_1v_1, m_2v_2 = momenta of body A and B after collision
According to Newton's third law:

$$F_1 = -F_2$$

$$\Rightarrow \frac{m_1v_1 - m_1u_1}{t} = -\frac{m_2v_2 - m_2u_2}{t}$$

$$\Rightarrow m_1v_1 - m_1u_1 = -m_2v_2 + m_2u_2$$

$$\Rightarrow m_1v_1 + m_2v_2 = m_1u_1 + m_2u_2$$

$$\Rightarrow \text{total momentum after collision} = \text{total momentum before collision}$$

Hence, momentum before and after collision is conserved.